

Denial of Service Attack Investigation

Summary: On January 2, 2026 at approximately 0800, the IT department received complaints of not being able to access the Cardinal Financial website. The scope of who and what was affected included customers, who were not able to access their financial data, and employees referencing the website for customer service. The security team reviewed the firewall logs and concluded a denial of service attack occurred, causing the web server to crash. The server was overloaded with requests from IP address 185.78.2.21 to the web server IP address of 192.168.3.100. The web server experienced a total of 45 minutes of downtime.

Response:

- A. Isolated the server by moving it to a sandbox segment of the network
- B. Resolved all connections to the server by clearing the cache
- C. Blocked the requesting IP address and created a network firewall rule=
 - a.

ACTION	SRC IP	DST IP	PROTOCOL	PORT
DENY	185.78.2.21	ANY	ALL	ANY

- D. Port 80 was closed on the server, only allowing secure traffic on port 443.
- E. Updated the server and applied the latest patch
- F. Team member used the Virus Total tool and concluded that the source IP address was highly malicious
- G. The host intrusion prevention system was configured with a new behavior analysis rule to block DOS attacks patterns that will detect deviations from baseline
- H. The server was scanned for additional vulnerabilities

Recovery:

- A. Before making the server live again, a team member made a connection to the server while it was still in the sandbox, to ensure it was running efficiently.
- B. Recommendation made to stakeholders to implement a backup webserver to decrease downtime.

Date	Time	Src IP	Dst IP	Src Port	Dst Port	Protocol	Action
2026-01-02	08:00:01	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:02	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:03	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:04	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW

2026-01-02	08:00:05	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:06	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:07	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:08	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:09	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:10	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:11	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:12	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:13	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:14	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:15	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:16	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:17	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:18	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW
2026-01-02	08:00:19	185.78.2.21	192.168.3.100	49152	80	TCP	ALLOW